}.

Assignment NO: 3

1)

```
1
1   2
1   2   3
1   2   3   4
1   2   3
1   2
1.
```

```
for (int i=1 ; i≤n ; i++) {
    for (int j=1 ; j≤n-i ; j++){
        print (" ");
    }
    for (int k=1 ; k≤i ; k++){
        print (k + " ");
    }
    S.opln();
for (int i= n-1 ; i≥1 ; i--){
    for (int j=1 ; j≤n-i ; j++)
        print (" ");
    }
    for (int k=1 ; k≤i ; k++){
        print (k + " ");
}.
```

$o(n^2) + o(n) + o(n^2) + o(n)$

$o(2n^2) + o(2n)$

$o(n^2)$

2).

```
1 _ _ _ _ _ _ 1
1 2 _ _ _ _ 2 1
1 2 3 _ _ 3 2 1
1 2 3 4 4 3 2 1
1 2 3 4 4 3 2 1
1 2 3     3 2 1
1 2         2 1
1             1
```

n=4

```
for (int i=1; i≤n ; i++) {
    for (int j=1 ; j≤i ; j++) {
        s.o.p (j+" ");
}
for (int k=1 ; k≤2(n-i); k++) {
        s.o.p (" ");
}
for (int m=i ; m≥1 ; m--){
        s.o.p (m+" ");
}
for (int i=n ; i≥1 ; i--) {
    for (int j=1 ; j≤i ; j++) {
        s.o.p (j+" ");
}
for (int k=1 ; k≤2(n-i); k++) {
        s.o.p (" ");
}
for (int m=i ; m≥1 ; m--) {
        s.o.p (m+" ");
}
```

3)

n=5

(11) (12) (13) (14) (15)

1 1

(21) 2 (22) (23) (24) 2 (25)

(31) (32) 3 (33) (34) (35)

(41) (42) (43) 4 (44) 4 (45)

(51) 5 (52) (53) (54) 5 (55)

```
for (int i=1 ; i≤n ; i++) {
    for (int j=1 ; j≤n ; j++) {
        if (i==j || i+j = n+1) {
            print (i) ;
        } else {
            print (" ");
        }
    }
}
```

1

i=1 j=1 , 155	j=4 , 455	(+1) i=2 ; 255
1==1 / 1+1 = 6 ✓	1==4 / 1+4=6. ✗	j=1 ; 155
print (i).	print (" ");	2=1 / 2+1=6 ✗
j=2 ; 255	j=5 ; 555 ,	print (" ");
1==2 / 1+2=6 ✗.	1==5 / 1+5=6 ✓	j=2 ; 255
print (" ");	print (i)	2==2 / 2+2=6 ✓
j=3 , 355	j=6 ; 655 ✗	print (i)
1==3 / 1+3=6 ✗.		j=3 ; 355
print (" ");		2==3 / 2+3=6 ✗
		print (" ");

4)

```
   1
  2  3
 4  5  6
7  8  9  10
11  12  13
 14  15
   16
```

```
n = 4
count = 1 ;
for (int i=1 ; i≤n ; i++) {
    for (int j=1 ; j≤n-i ; j++) {
        print (" ");
    }
    for (int k=1 ; k≤i ; k++) {
        print (count);
        count++ ;
    }
    S.opln () ;
    for (int m = n-1 ; m ≥ i ; m--) {
        print (count) ;
    }
```

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5).

```
    1
  0 1
 1 0 1
0 1 0 1
 1 0 1
  0 1
    1
```

n = 4.

```
for (int i = 1 ; i ≤ n ; i++) {
    for (int j = 1 ; j ≤ n - i ; j++) {
        print (" ");
    }
    for (int k = 1 ; k ≤ i ; k++) {
        if ((i + j) % 2 == 0) {
            print (1);
        else {
            print (0);
        }
    }
```